SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I Balaji S A 192126014 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%	2	2	2	1
Identification of Key Issues	20%	2	2	1	1
Application of Concepts	25%	1	1	1	1
Decision Making	20%	1	1	1	1
Clarity of Response	15%	1	1	1	1

25/30

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
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		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

SIMATS ENGINEERING

ASSESSMENT TOOL 2 — COURSE SOLUTIONS

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COURSE OUTCOME: CO2 (BL3) **WEIGHTAGE:** 30%

CODE OF CONDUCT & DECLARATION

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the academic integrity guidelines specified for this assessment.

Signature of Student: _____ **Date:** ____ / ____ / _____

ASSESSMENT RUBRICS (WEIGHTAGE: 30%)

Criteria	Weight	Q1 (10M)	Q2 (10M)	Q3 (10M)	Q4 (10M)	Q5 (10M)
Understanding Scenario	20%					
Identification of Issues	20%					
Application of Concepts	25%					
Decision Making	20%					
Clarity of Response	15%					

QUESTION 1: Smart Campus Variable Length Subnet Allocation (VLSM)

A Smart Campus is allocated IP block `192.168.100.0/24`. The network administrator assigns subnets based on size: School of Computing (`/25`), School of Electronics (`/26`), and School of Mechanical Engineering (`/27`). Design the allocation and identify host ranges and unused space.

Comprehensive VLSM Allocation Plan:

Department / Subnet	Mask / Prefix	Network Address	Usable Host Range	Broadcast Address	Usable Hosts
Computing	`/25` (255.255.255.128)	`192.168.100.0`	`192.168.100.1 - 192.168.100.126`	`192.168.100.127`	126 (2^7 - 2)
Electronics	`/26` (255.255.255.192)	`192.168.100.128`	`192.168.100.129 - 192.168.100.190`	`192.168.100.191`	62 (2^6 - 2)
Mechanical	`/27` (255.255.255.224)	`192.168.100.192`	`192.168.100.193 - 192.168.100.222`	`192.168.100.223`	30 (2^5 - 2)

Remaining Unused IP Address Range:

The allocated block ends at `192.168.100.223`. The remaining unassigned space within `192.168.100.0/24` is:

• **Unused IP Range:** `192.168.100.224` to `192.168.100.255` (Block size: 32 addresses, equivalent to a free `/27` subnet).

QUESTION 2: Enterprise Headquarters Subnet Design

A multinational software company expands its headquarters with private IP block `10.10.0.0/16`. Allocated subnets: Software Development (`/24`), Quality Assurance (`/25`), Human Resources (`/26`), and Executive Management (`/27`). Perform the subnet allocation analysis.

Departmental Subnet Allocation Table:

Department	Prefix & Subnet Mask	Network Address	Valid Host IP Range	Broadcast Address	Usable Hosts
Software Dev	`/24` (255.255.255.0)	`10.10.0.0`	`10.10.0.1 - 10.10.0.254`	`10.10.0.255`	254 (2^8 - 2)
Quality Assurance	`/25` (255.255.255.128)	`10.10.1.0`	`10.10.1.1 - 10.10.1.126`	`10.10.1.127`	126 (2^7 - 2)
Human Resources	`/26` (255.255.255.192)	`10.10.1.128`	`10.10.1.129 - 10.10.1.190`	`10.10.1.191`	62 (2^6 - 2)
Executive Mgmt	`/27` (255.255.255.224)	`10.10.1.192`	`10.10.1.193 - 10.10.1.222`	`10.10.1.223`	30 (2^5 - 2)

Unused IP Range Remaining in 10.10.0.0/16 Block:
- Currently allocated range: `10.10.0.0` to `10.10.1.223`.
- Immediate contiguous unused block: `10.10.1.224` to `10.10.1.255`.
- Entire remaining block: `10.10.2.0` to `10.10.255.255` (Total unused capacity: 65,056 addresses).

QUESTION 3: Smart Hospital IPv6 Migration & Addressing

A smart hospital migrates to IPv6 with assigned prefix `2001:0db8:abcd:0000:0000:0000:0000:0000/64`. Perform IPv6 compression, expansion, prefix identification, and host capacity calculation.

1. Compressed IPv6 Notation:

Applying RFC 5952 compression rules (leading zeros omitted in a hextet and consecutive all-zero hextets replaced by double colon `::`):
- Full: `2001:0db8:abcd:0000:0000:0000:0000:0000`
- Compressed: `2001:db8:abcd::/64`

2. Expanded Hexadecimal Representation:

Expanding `::` back to 5 all-zero hextets and restoring leading zeros to 4 digits per block:
- Full 128-bit expression: `2001:0db8:abcd:0000:0000:0000:0000:0000`

3. Network Prefix (/64 Prefix Length):

The first 64 bits represent the network prefix: `2001:0db8:abcd:0000::/64`

4. Total Supported Host Addresses:

In a `/64` subnet, the host portion consists of 128 − 64 = 64 bits.
- **Total Host Addresses:** 2^{64} = **18,446,744,073,709,551,616** (18.44 quintillion unique addresses).
- *Note:* Unlike IPv4, standard IPv6 SLAAC auto-configuration utilizes all 2^{64} addresses without requiring network/broadcast subtractions.

QUESTION 4: ISP Core Router Longest Prefix Match Analysis

An ISP core router contains entries: `192.168.1.0/24`, `192.168.2.0/24`, and `192.168.3.0/24`. A packet arrives with destination IP `192.168.2.45`. Determine routing actions.

1. Destination Subnet Identification:
Applying a `/24` mask (`255.255.255.0`) to destination `192.168.2.45`:
• Binary AND: `192.168.2.45 AND 255.255.255.0 = 192.168.2.0`
• The packet belongs to the `192.168.2.0/24` subnet.

2. Matching Routing Table Entry:
The router matches the packet destination against routing table entries using Longest Prefix Matching (LPM).
• Matching entry: `192.168.2.0/24`

3. Next-Hop Network / Interface:
The packet will be forwarded directly to the **locally connected Interface associated with the `192.168.2.0/24` subnet** (e.g., `eth1` or `GigabitEthernet0/1`).

4. Default Route Configuration:
If an incoming packet does not match any prefix in the routing table, the router forwards it to the Default Route:
• Default Route entry: `0.0.0.0/0` (Gateway of Last Resort).

QUESTION 5: Layer-3 Inter-Subnet Routing & Host Configuration

Two office departments are connected via a Layer-3 Router:
• Admin LAN: `192.168.10.0/26` (Gateway: `192.168.10.1`)
• Finance LAN: `192.168.10.64/26` (Gateway: `192.168.10.65`)

1 & 2. Network Boundaries & Valid Ranges:

Department LAN	Network Address	Broadcast Address	Valid Host IP Range	Default Gateway
Administration LAN	`192.168.10.0`	`192.168.10.63`	`192.168.10.1 - 192.168.10.62`	`192.168.10.1`
Finance LAN	`192.168.10.64`	`192.168.10.127`	`192.168.10.65 - 192.168.10.126`	`192.168.10.65`

3. Suggested IP Assignments for 3 Computers per LAN:
Note: The Gateway IPs (.1 and .65) are excluded from PC assignments.
• **Administration LAN PCs:** `PC1: 192.168.10.2`, `PC2: 192.168.10.3`, `PC3: 192.168.10.4` (Subnet Mask: `255.255.255.192`)
• **Finance LAN PCs:** `PC1: 192.168.10.66`, `PC2: 192.168.10.67`, `PC3: 192.168.10.68` (Subnet Mask: `255.255.255.192`)

4. Default Gateway Identification:
• Administration LAN hosts must use `192.168.10.1`.
• Finance LAN hosts must use `192.168.10.65`.